

CSA1706-ARTIFICIAL INTELLIGENCE

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Constraint-Based Problem Solving

1. Hospital Doctor Scheduling using Backtracking Search

Problem Statement

Assign **3 doctors (D1, D2, D3)** to **3 shifts**:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work **Night**.
2. D2 must work **before** D3.
 - Morning < Afternoon < Night
3. Only **one doctor** per shift.
4. D3 cannot work **Morning**.
5. Every doctor must be assigned exactly one shift.

Variables

- D1
- D2
- D3

Domain

Each doctor can be assigned:

{Morning, Afternoon, Night}

Initial State

D1 = ?

D2 = ?

D3 = ?

Backtracking Search Steps

Step 1

Assign

D1 = Morning

Constraint Check

- D1 ≠ Night ✓
- No conflict ✓

Current Assignment

D1 = Morning

Step 2

Assign

D2 = Morning

Constraint Check

Two doctors in same shift ✗

Violation

Only one doctor per shift

Backtrack

Step 3

Try

D2 = Afternoon

Constraint Check

- Different shift ✓
- D2 before D3 ✓

Current Assignment

D1 = Morning

D2 = Afternoon

Step 4

Assign

D3 = Morning

Constraint Check

D3 cannot work Morning ✗

Backtrack

Step 5

Assign

D3 = Afternoon

Constraint Check

Already occupied by D2 ✗

Backtrack

Step 6

Assign

D3 = Night

Constraint Check

- D3 $\neq$ Morning ✓
- D2 before D3 ✓
- One doctor per shift ✓

Goal reached.

Search Tree

Start

D1=Morning

/ | \

D2=M D2=A D2=N

X | |

D3=M

X

D3=A

X

D3=N

Solution

Final Schedule

Doctor Shift

D1 Morning

D2 Afternoon

D3 Night

Final Answer

Morning → D1

Afternoon → D2

Night → D3

All constraints are satisfied.

2. Robot Navigation using BFS Search

Grid (5×5)

S . . X .

. X . X .

. X . . .

. . X X .

. . . . G

Legend

S = Start

G = Goal

X = Obstacle

. = Free cell

Movement

- Up
- Down
- Left
- Right

Cost = 1

Manhattan Distance Formula

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

Although BFS does not use heuristics for expansion, the Manhattan distance can be calculated to estimate closeness to the goal.

BFS Expansion

Queue

Initially

Queue

[S]

Expand S

Visited

(0,0)

Insert

(0,1)

(1,0)

Queue

[(0,1),(1,0)]

Expand (0,1)

Insert

(0,2)

Queue

[(1,0),(0,2)]

Expand (1,0)

Insert

(2,0)

Queue

[(0,2),(2,0)]

Expand (0,2)

Insert

(1,2)

Queue

[(2,0),(1,2)]

Continue expansion until Goal is reached.

Optimal Path

S

↓

↓

↓

→

→

↓

↓

→

→

G

Coordinates

(0,0)

↓

(1,0)

↓

(2,0)

↓

(3,0)

→

(3,1)

↓

(4,1)

→

(4,2)

→

(4,3)

→

(4,4)=Goal

Path Cost

Number of moves

8

Therefore

Optimal Cost = 8

3. Autonomous Rescue Robot

(a) Constraint Analysis

Constraint	Effect on Decision Making
Move Up/Down/Left/Right	Robot has only four possible actions.
Obstacles	Robot cannot move through blocked cells and must find alternate paths.
Limited sensing	Robot only knows adjacent cells and updates knowledge as it explores.
Movement cost = 1	Robot prefers shorter paths.

Constraint	Effect on Decision Making
Risk zone cost = +2	Robot avoids risky cells when possible.
Online search	Robot continuously updates its map and replans when new information is discovered.

Decision Process

Robot repeatedly performs:

1. Sense surroundings.
 2. Update map.
 3. Calculate possible moves.
 4. Avoid obstacles.
 5. Avoid risky zones if cheaper alternatives exist.
 6. Move toward the goal.
 7. Repeat until survivor is reached.
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(b) Solution Approach

Suitable Search Strategy

Online Search Agent with Uniform Cost Search (UCS)

Steps

Step 1

Robot starts at

S

Step 2

Sense neighbouring cells.

Step 3

Update internal map.

Step 4

Insert reachable cells into a priority queue according to cumulative cost.

Step 5

Expand the node with the smallest total cost.

Step 6

If a new obstacle appears

Update the map and recompute the least-cost path.

Step 7

Continue until

Goal (G)

is reached.

Pseudo Code

Start

Insert Start into Priority Queue

While Goal not reached

 Sense adjacent cells

Update Map

Expand lowest-cost node

Ignore blocked cells

Add movement cost

Add risk cost if required

Insert new nodes

End While

Return Goal Path

(c) AI Concepts

1. Intelligent Agent

The robot

- Perceives the environment.
 - Processes information.
 - Chooses the best action.
 - Learns new obstacles.
 - Updates its plan.
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2. Rationality

The robot is rational because it always chooses the action with the **lowest total cost** while ensuring safety.

3. Environment Type

Property	Type
Observable	Partially Observable
Agents	Single Agent
Deterministic	Stochastic
Sequential	Yes
Dynamic	Yes
Discrete	Yes
Known	Partially Known

(d) Justification

The **Online Search Agent combined with Uniform Cost Search (UCS)** is the most suitable approach because:

- It can handle **unknown and changing environments**.
- It dynamically updates its map as new information becomes available.
- UCS guarantees the **least-cost path** by considering movement and risk costs.
- It avoids blocked cells and minimizes total travel cost.
- It supports intelligent decision-making in partially observable environments.

This approach ensures that the rescue robot reaches the survivor safely, efficiently, and with the minimum possible cost.

Summary

Question	Method Used	Final Result
1. Hospital Scheduling	Backtracking Search	Morning → D1, Afternoon → D2, Night → D3
2. Robot Navigation	Breadth-First Search (BFS)	Optimal path found with total cost = 8
3. Rescue Robot	Online Search Agent + Uniform Cost Search	Safely reaches the survivor with minimum total cost while adapting to new information